ENGINEER
a world of
DIFFERENCE

WALTER SCOTT, JR.
COLLEGE OF ENGINEERING
COLORADO STATE UNIVERSITY

Chemical and Biological Engineering Teaching Lab

ABOUT THE SPACE:

Chemical and biological engineers (CBE) turn raw materials into the products we rely on every day, from medicines and fuels to food and plastics, by designing the processes that make them safely and at scale. The CBE Teaching Lab is where students put those processes into practice.

Taken in the fourth year, the labs in this space teach chemical and biological processes, material testing techniques, fluid flow behavior, and separations. In this lab, CBE students use:

- Bioreactors: These machines precisely control conditions like temperature, pressure, and mixing to grow living cells. Students grow E. coli, optimize how much oxygen the bacteria take up, and analyze the products made. In industry, bioreactors are used to brew beer, make vaccines, insulin, and targeted chemotherapy, and produce biofuels and bioplastics as a sustainable alternative to traditional chemical plants.
- Membrane separation units: These selectively transfer oxygen from one stream to another. A large-scale gas separator can fill shipping crates with only nitrogen, which slows the ripening of fruit and allows places like Colorado to get fresh tropical produce.
- Hemodialysis: Oxygen can also be transferred from a gas into a liquid. This is how a heart-lung machine works, bypassing the lungs to supply oxygen to the blood during open-heart surgery. Students use water instead of blood to study flow rates, efficiency, and varying pressures.